

OpenClaw Temperature and Humidity Acquisition Tutorial

Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on DHT-related content.

1. Tutorial Objectives

This article demonstrates OpenClaw's natural language control capabilities through the DHT temperature and humidity sensor. After completing this tutorial, readers can:

- Understand the basic approach to reading temperature and humidity with OpenClaw
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four DHT cases: single reading, comfort judgment, threshold alert, OLED linked display
- Understand the complete chain from "user question" to "sensor reading" and "result feedback"

2. Hardware Preparation

- OpenClaw mainboard
- DHT temperature and humidity sensor [Default DHT11]
- [Optional] OLED display
- [Optional] Microphone module
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Low-level driver code: `/home/jetson/hardware_ha1_demo/`
- Temperature and humidity reading command: `/home/jetson/hardware_ha1_demo/dht`
- Temperature and humidity linked display: `/home/jetson/hardware_ha1_demo/dht_oled`

- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The `dht` tool already supports the following capabilities:

- Read current temperature and humidity
- Output results in `json` or `text` format
- Use `mock` fake data for local debugging

```
dht_oled #Display temperature and humidity data to OLED
dht #Feedback `json` results
```

If you want to display temperature and humidity results directly to OLED, you can also use `dht_oled`.

Note: If you want to write the tutorial yourself to read temperature and humidity data from the `ajetson` in the Python driver library, it is recommended to read DHT data at 2-second intervals, because DHT sensor itself is more suitable for 2s sampling

4. Test Temperature and Humidity Acquisition

Before starting this tutorial, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After confirming OpenClaw can respond normally, enter DHT-specific testing.

Terminal input to enter TUI:

```
openclaw tui
```

```

[11:11] OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws--type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens 0/1.0m (0%)

```

You can try entering the following content in the terminal:

- Read current temperature and humidity
- What is the current environmental humidity
- Help me judge if the current environment is comfortable
- Remind me if the temperature is too high

```
Read current temperature and humidity

Current temperature and humidity readings:
- Temperature: 29.9°C (85.8°F)
- Humidity: 52%

Sensor: DHT11 ✓

What is the current environmental humidity

Current environmental humidity: 52%

(Temperature is 29.9°C if you need that too) 01F
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
48k/1.0m (5%)
```

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote control, classroom demonstrations	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the control chain
Voice code	Voice interaction, complete AI experience	Closest to real conversation experience	As final demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

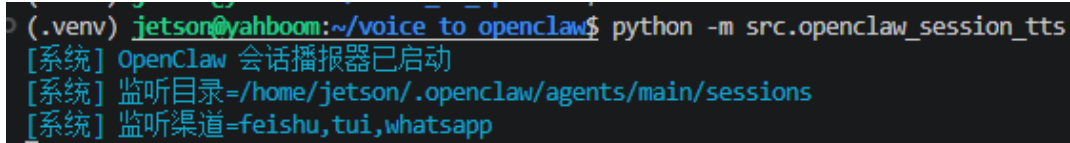
- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

ENABLE_TTS=true #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



A terminal window showing the execution of the program. The prompt is (.venv) jetson@yahboom:~/voice to openclaw\$. The command python -m src.openclaw_session_tts has been entered. The output shows three system messages in Chinese: [系统] OpenClaw 会话播报器已启动, [系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions, and [系统] 监听渠道=feishu,tui,whatsapp.

```
(.venv) jetson@yahboom:~/voice to openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Solution

For whatsapp connection steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. This section only retains dialogue examples suitable for the DHT tutorial.

After creation, you can directly dialogue, for example:

- Read current temperature and humidity
- Judge if it's currently too hot
- Remind me if humidity is too low

19:53



< 2



+86 153 3885 7526 (你)

给自己发消息

OLED

19:52 ✓✓

Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.

19:52 ✓✓

Read current temperature and humidity

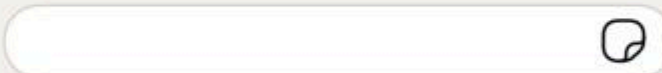
19:53 ✓✓

Current readings from the DHT11 sensor:

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 52%

The environment is fairly warm and moderately humid.

19:53 ✓✓



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123



space

return



5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter the following command in the terminal to continue the dialogue:

```
openclaw tui
```

```
OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws—type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_user_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration].

You need to say "Hi, yahboom" to wake up for dialogue.

After waking up, you can directly say control content, for example:

- What is the current temperature
- Help me read the humidity
- Tell me if it's too stuffy

```

[系统] ASK=xmfc1_ws/1dc/ch_us | 发布=gateway_chat | TTS=ji / xmfc1_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Read current temperature and humidity.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Current readings from the DHT11 sensor:

- **Temperature:** 30°C
  (86°F)
- **Humidity:** 51%

The sensor is working well! 🌡️💧
[信息] 回复摘要失败, 已退回本地清洗播报
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6810 ms 语音, 正在转写

[你] It's too hot. Please, let me know.
[发送] 正在交给 OpenClaw

```

6. Comprehensive Cases

Note: The following are all dialogue demonstrations using `openclaw_tui` method. You can also choose whatsapp or voice method to complete.

6.1 Case One: Single Reading

Experimental Objective

Get current temperature and humidity once to complete the most basic environmental sampling.

Effect Description

After receiving the command, OpenClaw will call the DHT driver to read the current temperature and humidity, and return the results to the user in natural language.

Example Instructions

- Read current temperature and humidity
- What is the current indoor temperature
- Help me check the current humidity

6.2 Case Two: Comfort Judgment

Experimental Objective

Let OpenClaw make a simple judgment about the current environment after reading data.

Effect Description

Besides directly reporting numbers, OpenClaw can also roughly judge states like "too hot", "too cold", "too dry", "too stuffy" based on temperature and humidity, making results closer to natural dialogue.

Example Instructions

- Help me judge if the current environment is comfortable
- Is the current temperature and humidity suitable for long-term office work
- Give me a brief evaluation based on temperature and humidity

6.3 Case Three: Threshold Alert

Experimental Objective

Let OpenClaw give alerts when temperature or humidity exceeds the set range.

Effect Description

When temperature is too high, too low, or humidity is too high or too low, OpenClaw can directly output alert information, or link with RGB lights for more obvious feedback.

Example Instructions

- Remind me if temperature exceeds 30 degrees
- Prompt if humidity is below 40% that air is too dry
- Are there any alarm conditions now

Extensible Gameplay

- Link with RGB for red, yellow, blue three-color alerts
- Link with OLED to display alarm text
- Add buzzer for stronger alerts

6.4 Case Four: OLED Linked Display

Experimental Objective

Display the temperature and humidity results just read directly to OLED, forming a complete data display chain.

Effect Description

After completing an environmental sampling, OpenClaw can not only return results to the user, but also simultaneously write temperature and humidity to the OLED screen, forming a complete effect of "reading + display".

Example Instructions

- Read current temperature and humidity and display to OLED
- Write temperature and humidity to the screen
- Make a simple temperature and humidity display page

Extensible Gameplay

- Periodically refresh OLED content
- Make into a temperature and humidity dashboard
- Link with weather station project to form a complete application